Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/27/2004
PI Name:	Megaridis, Constantine M. Ph.D.		
Project Title:	Molten-Metal Droplet Deposition on a Moving Substrate in Microgravity: Aiding the Development of Novel Technologies for Microelectronic Assembly		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cmm@uic.edu	**Fax:**	FY 312 413-0447
PI Organization Type:	UNIVERSITY	**Phone:**	312 996-3436
Organization Name:	University of Illinois at Chicago		
PI Address 1:	Dept of Mechanical and Industrial Engineering (M/C 251)		
PI Address 2:	842 West Taylor Street		
PI Web Page:	http://www.mie.uic.edu/		
City:	Chicago	**State:**	IL
Zip Code:	60607-7022	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	07/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
Contact Email:	walter.m.duval@nasa.gov		
Flight Program:			
Flight Assignment:	note: extended 1 year		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Poulikakos, Dimos (Swiss Federal Inst. of Technology (ETH Zurich)) Nayagam, Vedha (NCMR)		
Grant/Contract No.:	NAG3-2456		
Performance Goal No.:	1.23		
Performance Goal Text:	Pursue ground-breaking basic research in fluid physics and transport phenomena to provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.		

Task Description:	The study consists of a theoretical and an experimental component. The theoretical work uses computer models based on finite element techniques. The experimental work is performed in microgravity in order to allow for the use of larger droplets that make feasible the performance of accurate measurements while keeping the effect of gravity negligible. The objective of this program is to investigate the fluid dynamics and simultaneous solidification of molten solder droplets impacting obliquely on a flat surface. The problem is directly relevant to the printing of microscopic solder droplets during surface mounting of microelectronic devices. The study is pursued in collaboration with an industrial partner, MicroFab Technologies, and seeks to identify the degree of influence of the dominant process parameters on specific aspects of solder droplet dispensing. The data produced is analyzed to define the domains of operational parameters that result in optimal and consistent production of solder bumps, as needed for specific industrial applications. The technology of solder microdroplet dispensing has shown good promise in microelectronic packaging and assembly. Therefore, the development of a good understanding of the pertinent fluid dynamics and solidification phenomena is essential for the successful commercial implementation of the technique. Conducting experiments in microgravity eliminates the unwanted influence of gravity and makes the experimental investigation of large droplet dispension directly relevant to the actual application, where solder droplets are approximately four thousands of an inch in diameter. This program advances the state of knowledge and provides a science base for the fundamental transport phenomena occurring in the solder microdroplet dispensing technology. The study also seeks to produce predictive tools to be used for the successful development of this innovative technique in electronic component manufacturing or other relevant technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Experiments were conducted to elucidate the contact angle dynamics of droplets striking orthogonally smooth surfaces with widely disparate wetting characteristics (wetting to non-wetting). The impact conditions corresponded to Re = O(100) - O(1000), We = O(1) -O(10), Ca = O(0.001) - O(0.01), Oh = O(0.001) and Bo = O(0.1). In this parameter regime, inertial, viscous, and capillary phenomena act simultaneously to influence contact line motion and arrest. Fundamental information on the relation between apparent (macroscopic) contact angle and contact line speed was obtained. For wetting configurations, the classical contact angle vs. contact line speed trend was observed, while for partially wetting configurations, advancing contact angles were found to be insensitive to contact line speed. Impact on non-wetting surfaces was followed by rebound, and revealed that both advancing and receding contact angles did not change with contact line speed. The data also provided clear evidence of contact angle hysteresis. It is concluded that droplet impact at moderate Reynolds numbers on surfaces of varying wetting characteristics does not support a simple relation between apparent contact angle and velocity.
Bibliography Type:	Description: (Last Updated: 06/10/2005)
Articles in Peer-reviewed Journals	Megaridis, C. M.; Boomsma, K.; Bayer, I. S. "Partial Rebound of Molten-Metal Droplets Impacting on Solid Substrates" AIChE J. 50: 1356-1363 , Jul-2004
Papers from Meeting Proceedings	Bayer, I. S.; Megaridis, C. M. "Contact Angle Dynamics of Droplets Impacting on Flat Substrates" XXI International Congress of Theoretical and Applied Mechanics, Warsaw, Poland Aug-2004
Presentation	Bayer, I. S.; Megaridis, C. M. "Contact Angle Dynamics of Droplets Impacting on Flat Substrates" Fourth International Symposium on Contact Angle Wettability and Adhesion, Philadelphia, PA Jun-2004